Lctr 11: Fourier Series and LTI Systems — Frequency Response and Filtering

Rogelio Gracia Otalvaro

Spring 2026

Lctr 11: Fourier Series and LTI Systems — Frequency Response and Filtering

Spring 2026

Focus: Sections 3.8 – 3.11 (Pages 226 to 249)

11.1 Connecting the Fourier Series to LTI Systems

Why This Matters

In Lectures 9 and 10 we learned how to decompose periodic signals into sums of complex exponentials (the Fourier series). In Lecture 6 we learned that the output of an LTI system is found by convolution. Now we bring these ideas together: when the input to an LTI system is periodic, the output is also periodic, and each frequency component is modified individually by the system. This leads to the concept of **frequency response**—which is the foundation of **filtering**, one of the most important applications in signal processing. From the equalizer on your music player to noise-canceling headphones to radio tuners, frequency-based analysis is everywhere.

11.1.1 Review: The Key Building Blocks

From our previous lectures:

- **Eigenfunction property (Lecture 9):** When a complex exponential $e^{jk\omega_0 t}$ passes through a continuous-time LTI system, the output is $H(jk\omega_0)\, e^{jk\omega_0 t}$. The system multiplies the exponential by a complex constant.
- **Fourier series (Lectures 9–10):** Any periodic signal can be written as a sum of harmonically related complex exponentials: $x(t) = \sum_k a_k\, e^{jk\omega_0 t}$.
- **Superposition (Lecture 6):** For a linear system, the response to a sum of inputs is the sum of the individual responses.

Combining these three ideas gives us a powerful result.

Visual Intuition

The three-step logic: The figure below shows how the eigenfunction property, Fourier decomposition, and superposition combine. Each harmonic "passes through" the system independently.

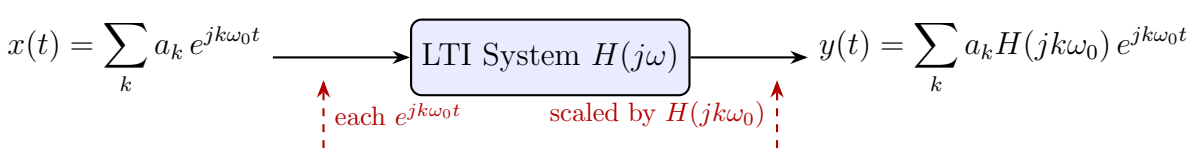

Figure 1: An LTI system processes each harmonic independently—this is the central idea.

11.2 The Frequency Response of an LTI System

11.2.1 Definition: Continuous-Time Frequency Response

For a continuous-time LTI system with impulse response $h(t)$, the **frequency response** is:

$$\boxed{H(j\omega) = \int_{-\infty}^{+\infty} h(\tau)\, e^{-j\omega\tau}\, d\tau}$$

This is a **function of frequency** ω. It tells us how the system modifies a complex exponential at frequency ω.

$H(j\omega)$ is complex-valued, so it has both magnitude and phase:

- $|H(j\omega)|$ = how much the system **amplifies or attenuates** the signal at frequency ω.
- $\angle H(j\omega)$ = how much the system **shifts the phase** of the signal at frequency ω.

Visual Intuition

The plots below show what a typical frequency response *looks* like. The magnitude plot tells you "how loud" each frequency comes out; the phase plot tells you "how delayed" each frequency is.

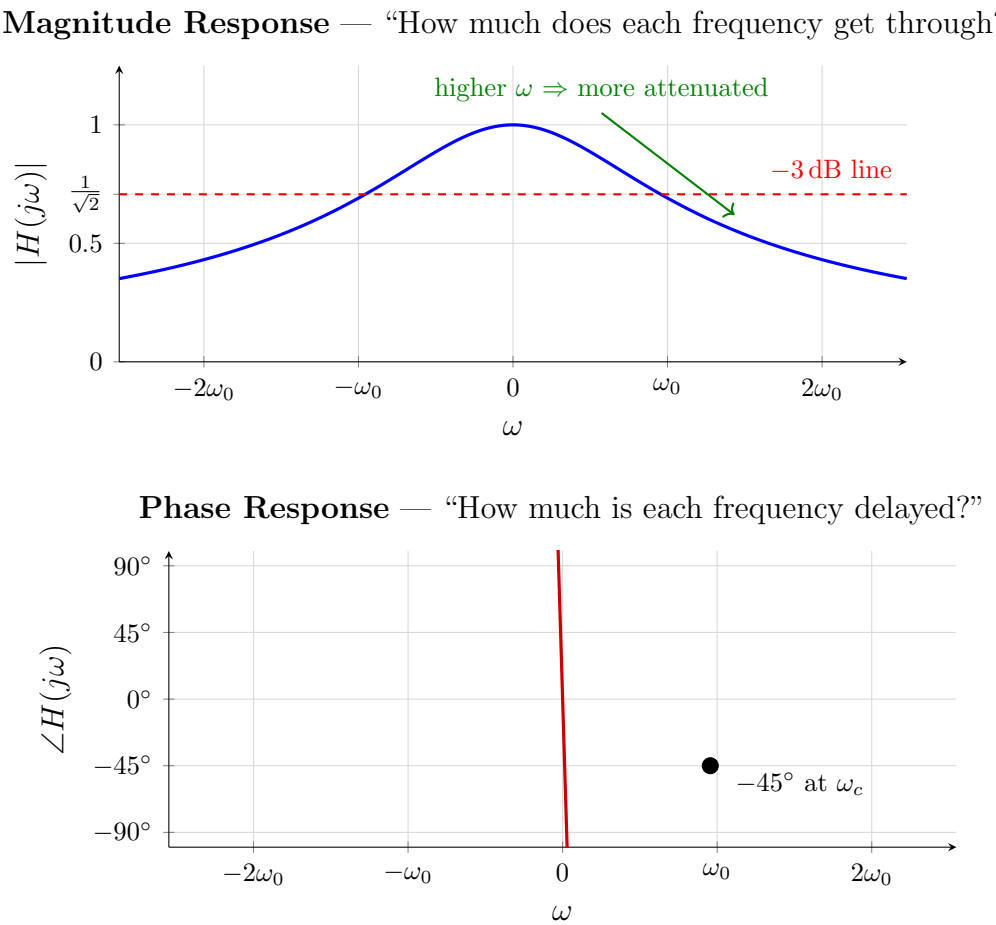

Figure 2: Generic magnitude and phase response of a lowpass-type system (here $H(j\omega) = 1/(1+j\omega/3)$). The magnitude decreases for higher frequencies, and the phase becomes more negative.

11.2.2 Definition: Discrete-Time Frequency Response

For a discrete-time LTI system with impulse response $h[n]$:

$$\boxed{H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n]\, e^{-j\omega n}}$$

This is also a function of frequency ω, and it has the additional property that it is **periodic in ω with period 2π**. This is a direct consequence of the fact that $e^{j(\omega+2\pi)n} = e^{j\omega n}$ for integer n.

Key Insight

Key property of discrete-time frequency response: $H(e^{j\omega})$ repeats every 2π. This means that in discrete time, we only need to consider frequencies in one interval of length 2π (typically $-\pi < \omega \leq \pi$ or $0 \leq \omega < 2\pi$). Low frequencies are near $\omega = 0$, and high frequencies are near $\omega = \pm\pi$.

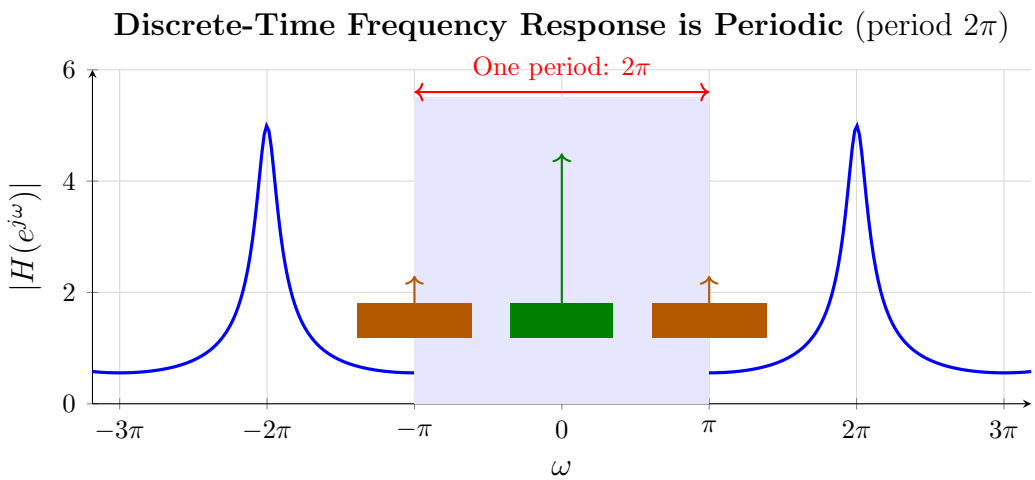

Figure 3: The DT frequency response $|H(e^{j\omega})|$ for $H(e^{j\omega}) = 1/(1 - 0.8e^{-j\omega})$ repeats every 2π. The shaded region is typically the only portion we analyze.

11.3 Output of an LTI System for a Periodic Input

11.3.1 Continuous-Time Case

If the input to a CT LTI system is the periodic signal:

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k \, e^{jk\omega_0 t}$$

then by superposition and the eigenfunction property, the output is:

$$\boxed{y(t) = \sum_{k=-\infty}^{+\infty} a_k \, H(jk\omega_0) \, e^{jk\omega_0 t}}$$

What this says: The output is also periodic with the *same* period, and its kth Fourier coefficient is:

$$b_k = a_k \cdot H(jk\omega_0)$$

Each input Fourier coefficient a_k is multiplied by the value of the frequency response at the corresponding harmonic frequency $k\omega_0$.

Visual Intuition

Think of the frequency response as a "recipe card" that the system applies to each harmonic independently. The figure below shows this multiplication visually.

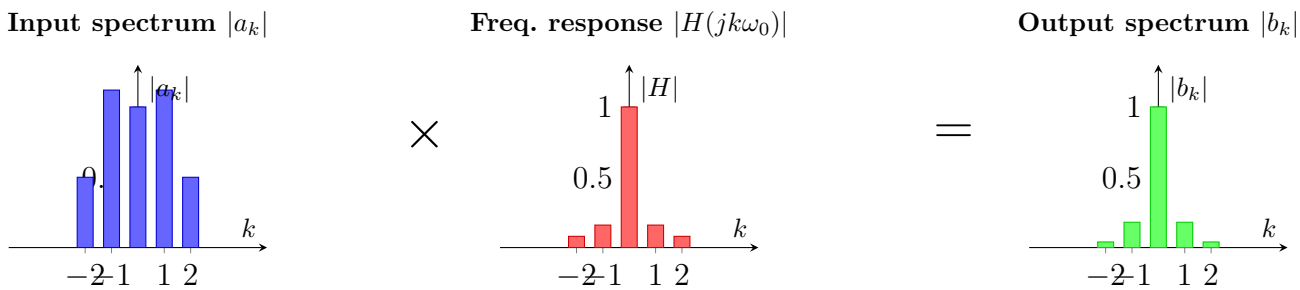

Figure 4: Frequency-by-frequency multiplication: $|b_k| = |a_k| \cdot |H(jk\omega_0)|$. The lowpass system strongly attenuates the higher harmonics ($k = \pm 2$), preserving the DC and fundamental while smoothing out the signal.

11.3.2 Discrete-Time Case

Exactly the same idea applies. If the input is:

$$x[n] = \sum_{k=\langle N \rangle} a_k \, e^{jk(2\pi/N)n}$$

then the output is:

$$\boxed{y[n] = \sum_{k=\langle N \rangle} a_k \, H\left(e^{j2\pi k/N}\right) e^{jk(2\pi/N)n}}$$

The kth output coefficient is $b_k = a_k \cdot H(e^{j2\pi k/N})$.

Key Insight

The big picture: An LTI system acts as a **frequency-by-frequency multiplier**. It independently scales and phase-shifts each harmonic component of the input. The frequency response $H(j\omega)$ (or $H(e^{j\omega})$) tells you the "recipe" for how each frequency is treated.

11.4 Worked Example: CT System with Periodic Input

Problem: Consider an LTI system with impulse response $h(t) = e^{-t}\,u(t)$ (where $u(t)$ is the unit step function). The input is the periodic signal from Lecture 9, Example 9.5.2:

$$x(t) = 1 + \sin(\omega_0 t) + 2\cos(\omega_0 t) + \cos(2\omega_0 t + \pi/4), \quad \omega_0 = 2\pi$$

Step 1: Compute the frequency response.

$$H(j\omega) = \int_0^\infty e^{-\tau}\, e^{-j\omega\tau}\, d\tau = \int_0^\infty e^{-(1+j\omega)\tau}\, d\tau = \frac{1}{1+j\omega}$$

Step-by-step: Evaluating the integral

We combine the two exponentials into one: $e^{-\tau} \cdot e^{-j\omega\tau} = e^{-(1+j\omega)\tau}$. Then:

$$\int_0^\infty e^{-(1+j\omega)\tau}\, d\tau = \left[\frac{e^{-(1+j\omega)\tau}}{-(1+j\omega)}\right]_0^\infty = 0 - \frac{1}{-(1+j\omega)} = \frac{1}{1+j\omega}$$

The integral converges because $\mathrm{Re}(1+j\omega) = 1 > 0$, so $e^{-(1+j\omega)\tau} \to 0$ as $\tau \to \infty$.

Step 2: Recall the Fourier coefficients of $x(t)$ from Lecture 9:

$$a_0 = 1, \quad a_1 = 1 - \frac{j}{2}, \quad a_{-1} = 1 + \frac{j}{2}, \quad a_2 = \frac{1}{2}e^{j\pi/4}, \quad a_{-2} = \frac{1}{2}e^{-j\pi/4}$$

Step 3: Compute the output coefficients $b_k = a_k \cdot H(jk\omega_0)$:

$$\begin{aligned}
b_0 &= a_0 \cdot H(j \cdot 0) = 1 \cdot \frac{1}{1+0} = 1 \\
b_1 &= a_1 \cdot H(j\omega_0) = \left(1 - \frac{j}{2}\right) \cdot \frac{1}{1+j2\pi} \\
b_{-1} &= a_{-1} \cdot H(-j\omega_0) = \left(1 + \frac{j}{2}\right) \cdot \frac{1}{1-j2\pi} \\
b_2 &= a_2 \cdot H(j2\omega_0) = \frac{1}{2}e^{j\pi/4} \cdot \frac{1}{1+j4\pi} \\
b_{-2} &= a_{-2} \cdot H(-j2\omega_0) = \frac{1}{2}e^{-j\pi/4} \cdot \frac{1}{1-j4\pi}
\end{aligned}$$

Step 4: Write the output:

$$y(t) = \sum_{k=-2}^{2} b_k\, e^{jk\omega_0 t}$$

Notice how *each* harmonic is affected differently: the system attenuates higher harmonics more because $|H(jk\omega_0)| = 1/\sqrt{1+(k\omega_0)^2}$ decreases as $|k|$ increases. This is the behavior of a **lowpass filter**.

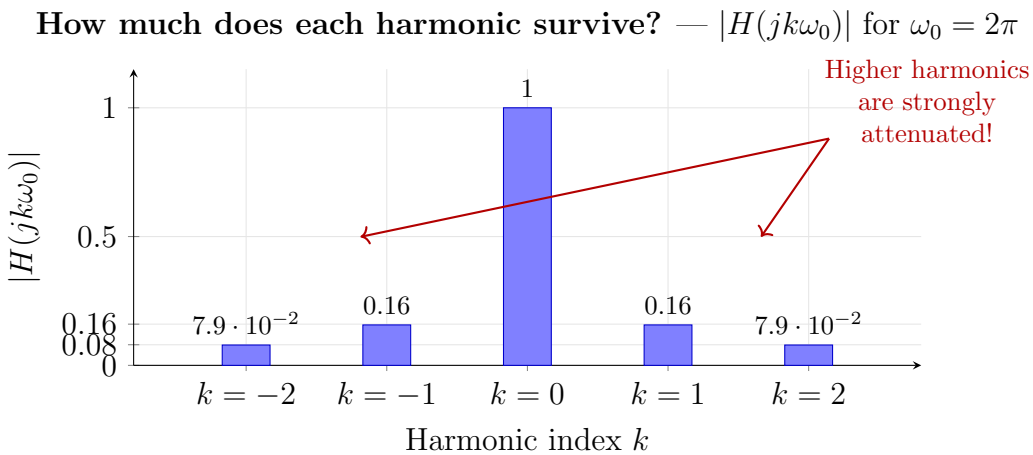

Figure 5: The magnitude $|H(jk\omega_0)|$ at each harmonic frequency. The DC component ($k = 0$) passes through unchanged, but the 1st and 2nd harmonics are attenuated to about 16% and 8%, respectively. This system is a very aggressive lowpass filter at $\omega_0 = 2\pi$.

11.5 Introduction to Filtering

Why This Matters

Filtering is the process of modifying the frequency content of a signal. Since the output Fourier coefficients are $b_k = a_k \cdot H(jk\omega_0)$, the frequency response $H(j\omega)$ determines which frequencies are passed and which are attenuated. By designing $H(j\omega)$ appropriately, we can build systems that perform specific frequency manipulations—like removing noise, separating radio channels, or equalizing audio.

11.5.1 Frequency-Shaping Filters

A **frequency-shaping filter** changes the relative amplitudes of different frequency components. Examples include:

- **Audio equalizers:** Boost bass (low frequencies) or treble (high frequencies) in music.
- **Differentiators:** The system $y(t) = dx(t)/dt$ has frequency response $H(j\omega) = j\omega$. It amplifies high frequencies and attenuates low ones, which is why differentiation enhances edges in images.

11.5.2 Frequency-Selective Filters

A **frequency-selective filter** aims to pass certain frequency bands and reject others. There are three main types:

1. **Lowpass filter:** Passes frequencies below a cutoff frequency ω_c and rejects those above.

$$H_{\text{ideal}}(j\omega) = \begin{cases} 1, & |\omega| \leq \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

2. **Highpass filter:** Passes frequencies above ω_c and rejects those below.

$$H_{\text{ideal}}(j\omega) = \begin{cases} 0, & |\omega| < \omega_c \\ 1, & |\omega| \geq \omega_c \end{cases}$$

3. **Bandpass filter:** Passes frequencies within a band $[\omega_{c1}, \omega_{c2}]$ and rejects all others.

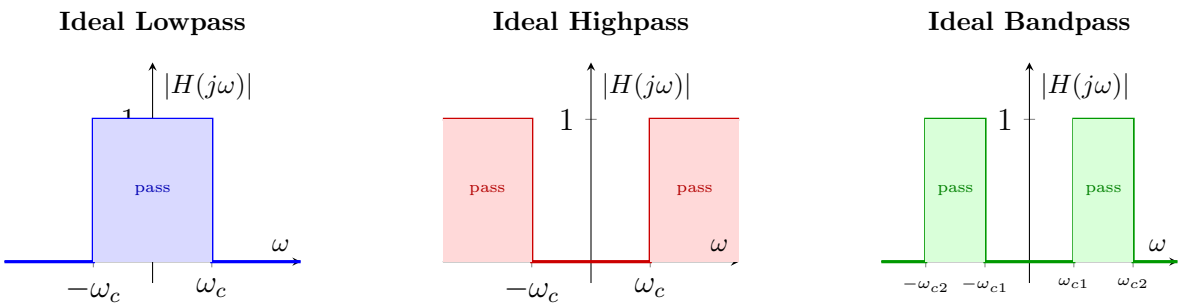

Figure 6: The three main types of ideal frequency-selective filters. In practice, the transitions are never perfectly sharp—they always have a gradual roll-off.

Terminology:

- **Passband:** The range of frequencies that the filter passes (approximately) without change.
- **Stopband:** The range of frequencies that the filter rejects (attenuates to approximately zero).
- **Cutoff frequency (ω_c):** The boundary between the passband and the stopband.

Common Mistake

Ideal vs. real filters: Ideal frequency-selective filters (with perfectly sharp transitions) **cannot be built** in practice. Real filters always have a gradual transition between the passband and the stopband. All the examples below are *non-ideal* approximations.

11.6 Example: A Simple RC Lowpass Filter

Consider the first-order RC circuit (Resistor–Capacitor circuit), where the input is the source voltage $v_s(t)$ and the output is the capacitor voltage $v_c(t)$. The system is described by the differential equation:

$$RC\,\frac{dv_c(t)}{dt} + v_c(t) = v_s(t)$$

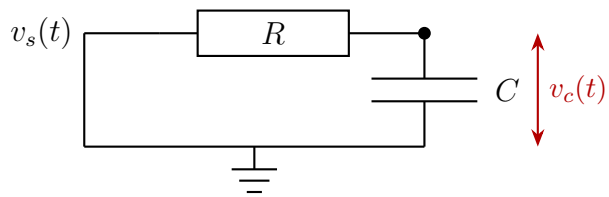

Figure 7: RC lowpass filter circuit. The output is taken across the capacitor.

11.6.1 Finding the Frequency Response

Substitute the eigenfunction: let $v_s(t) = e^{j\omega t}$, which means $v_c(t) = H(j\omega)\,e^{j\omega t}$.

Substituting into the differential equation:

$$RC \cdot j\omega\, H(j\omega)\, e^{j\omega t} + H(j\omega)\, e^{j\omega t} = e^{j\omega t}$$

Dividing both sides by $e^{j\omega t}$:

$$H(j\omega)(1 + j\omega RC) = 1$$

Solving:

$$\boxed{H(j\omega) = \frac{1}{1 + j\omega RC}}$$

11.6.2 Analyzing the Frequency Response

The magnitude and phase are:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}, \qquad \angle H(j\omega) = -\arctan(\omega RC)$$

Key observations:

- At $\omega = 0$ (DC): $|H(j\omega)| = 1$. Low frequencies pass through without attenuation.
- As $|\omega|$ increases: $|H(j\omega)|$ decreases toward 0. Higher frequencies are progressively attenuated.
- The **cutoff frequency** is approximately $\omega_c = 1/(RC)$. At this frequency, $|H(j\omega_c)| = 1/\sqrt{2} \approx 0.707$ (the signal power is halved, which corresponds to a drop of about 3 decibels, abbreviated dB).

This confirms: the RC circuit with capacitor voltage as output acts as a **lowpass filter**.

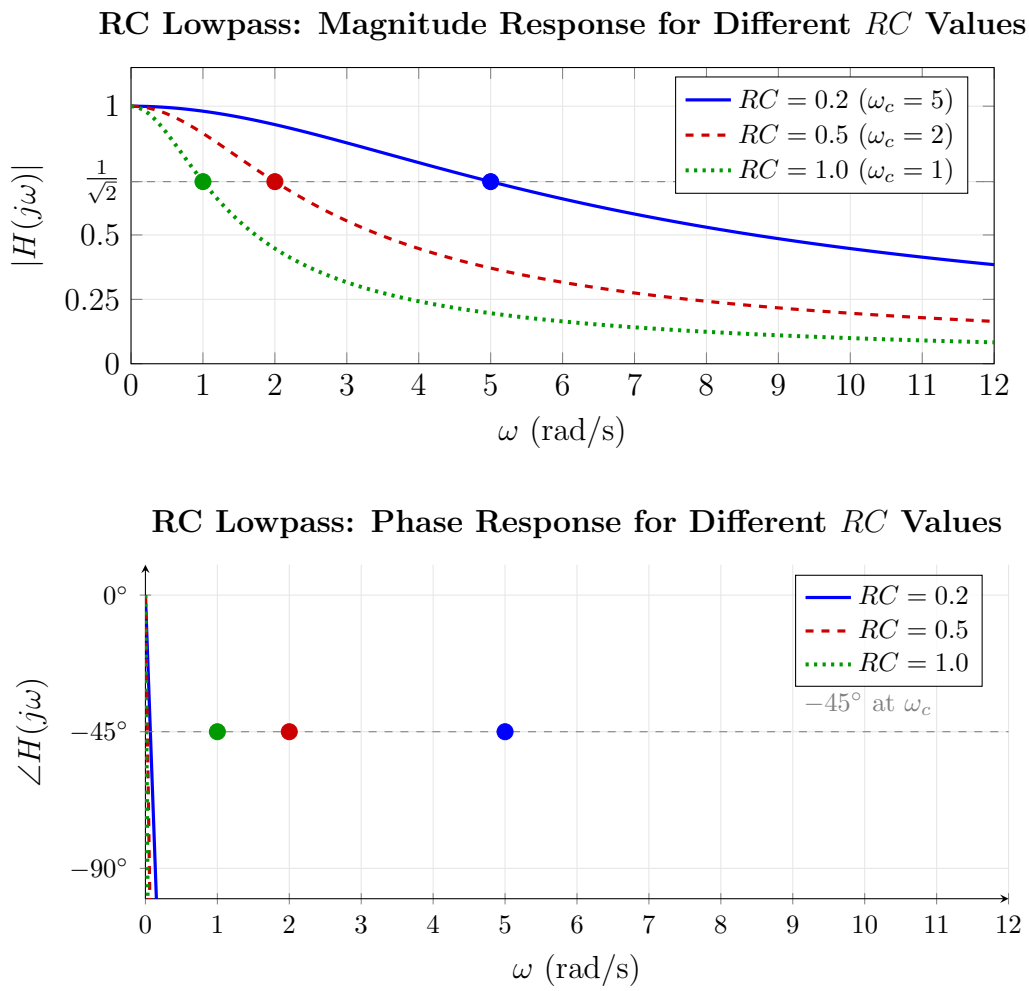

Figure 8: Larger RC means a lower cutoff frequency $\omega_c = 1/RC$: the filter rejects more frequencies. At the cutoff, the magnitude is $1/\sqrt{2}$ and the phase is $-45°$.

11.6.3 Time-Domain vs. Frequency-Domain Trade-off

The impulse response is $h(t) = \frac{1}{RC}\, e^{-t/RC}\, u(t)$ and the step response is $s(t) = (1 - e^{-t/RC})\, u(t)$.

Trade-off: If we want a sharper lowpass filter (smaller ω_c, meaning we reject more high frequencies), we need a *larger* RC. But a larger RC makes the step response slower—the system takes longer to respond to sudden changes. This time–frequency trade-off is fundamental and appears throughout signal processing and control systems.

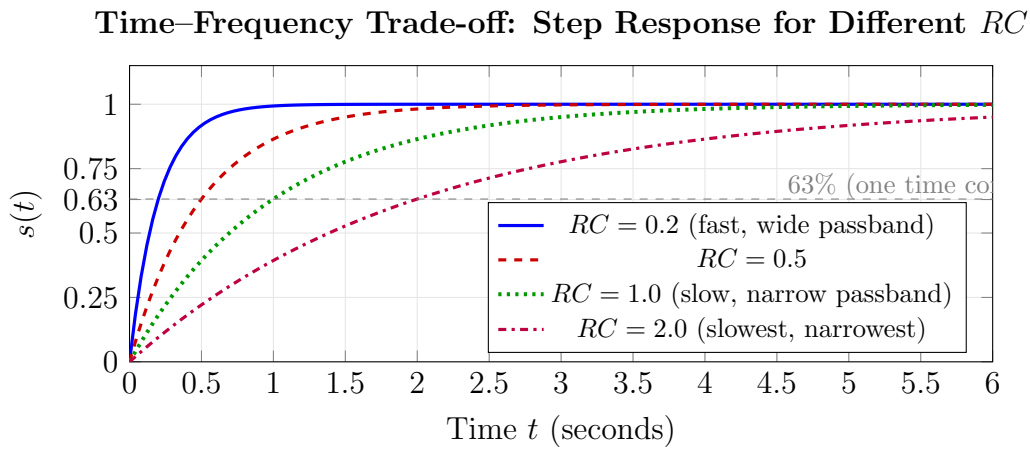

Figure 9: The fundamental trade-off: a tighter lowpass filter (larger RC) rejects more high frequencies but responds *slower* to sudden changes. You cannot have both sharp frequency selectivity and fast time-domain response—this is a universal principle.

11.7 Example: A Simple RC Highpass Filter

Using the same RC circuit but taking the *resistor voltage* $v_R(t)$ as the output instead:

$$G(j\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

Key observations:

- At $\omega = 0$: $|G(j\omega)| = 0$. The DC component is completely blocked.
- As $|\omega| \to \infty$: $|G(j\omega)| \to 1$. High frequencies pass through.

This is a **highpass filter**—from the same physical circuit, just by choosing a different output variable.

Visual Intuition

The same RC circuit produces *both* a lowpass and a highpass filter depending on where you measure the output! Notice that $|H(j\omega)|^2 + |G(j\omega)|^2 = 1$ at every frequency—the total power is conserved.

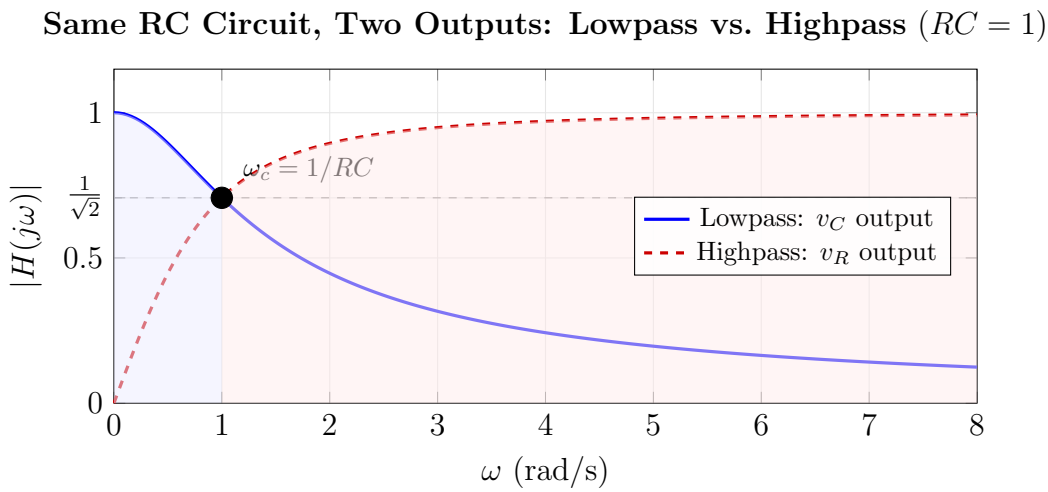

Figure 10: Both filters share the same cutoff frequency $\omega_c = 1/RC$. At the cutoff, both have magnitude $1/\sqrt{2}$. Note: $|H|^2 + |G|^2 = 1$ (conservation of energy).

11.8 Discrete-Time Filtering Examples

11.8.1 First-Order Recursive Filter

The difference equation:

$$y[n] - a\,y[n-1] = x[n], \qquad |a| < 1 \text{ (stability requirement)}$$

has frequency response:

$$H(e^{j\omega}) = \frac{1}{1 - a\,e^{-j\omega}}$$

Behavior:

- If $0 < a < 1$: The filter acts as a **lowpass filter**. It passes low frequencies (near $\omega = 0$) and attenuates high frequencies (near $\omega = \pi$).
- If $-1 < a < 0$: The filter acts as a **highpass filter**. It passes high frequencies and attenuates low ones.
- The parameter $|a|$ controls the sharpness: values closer to 1 give a narrower passband.

The impulse response is $h[n] = a^n u[n]$, which decays geometrically. Larger $|a|$ means slower decay (more memory) and a narrower passband.

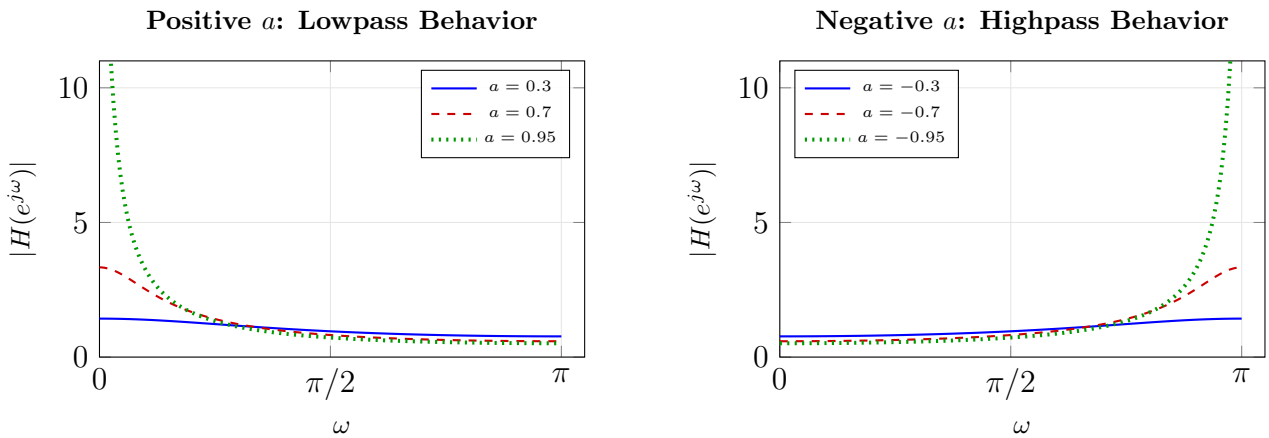

Figure 11: DT first-order recursive filter. **Left:** Positive a gives lowpass behavior (peak at $\omega = 0$). **Right:** Negative a gives highpass behavior (peak at $\omega = \pi$). As $|a| \to 1$, the peak becomes sharper but the system approaches instability.

Common Mistake

When $|a|$ is very close to 1 (e.g., $a = 0.99$), the filter becomes very frequency-selective *but* it is nearly unstable. In practice, numerical precision limits how close to 1 you can push $|a|$ before the filter "rings" excessively or becomes unpredictable. This is the discrete-time version of the time–frequency trade-off.

11.8.2 Moving-Average (Nonrecursive) Filters

A simple nonrecursive lowpass filter takes the average of neighboring input samples:

$$y[n] = \frac{1}{M+N+1} \sum_{k=-N}^{M} x[n-k]$$

The impulse response is a rectangular pulse (all weights equal), and the frequency response involves a $\sin(x)/\sin(y)$ pattern.

Two-point average example:

$$y[n] = \frac{1}{2}(x[n] + x[n-1]), \qquad H(e^{j\omega}) = e^{-j\omega/2}\cos(\omega/2)$$

This passes DC perfectly ($H(e^{j0}) = 1$) and completely rejects the highest frequency ($H(e^{j\pi}) = 0$), confirming lowpass behavior.

Simple highpass example:

$$y[n] = \frac{x[n] - x[n-1]}{2}, \qquad H(e^{j\omega}) = je^{-j\omega/2}\sin(\omega/2)$$

This rejects DC ($H(e^{j0}) = 0$) and passes the highest frequency ($|H(e^{j\pi})| = 1$), confirming highpass behavior.

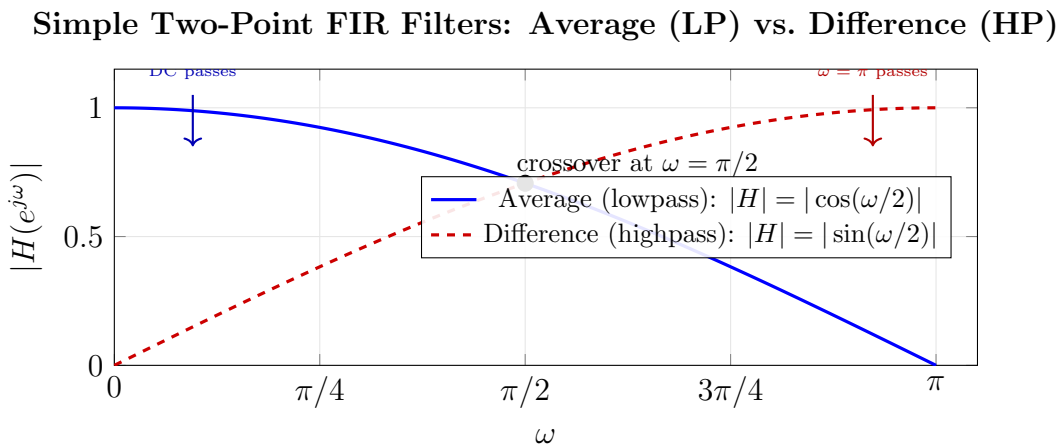

Figure 12: The simplest FIR lowpass and highpass filters. Note that both are *complementary*: when one passes a frequency, the other rejects it. These are the building blocks of more sophisticated filter designs.

Key Insight

Nonrecursive (FIR) filters have finite-length impulse responses and are **always stable** (since the impulse response is always absolutely summable). **Recursive (IIR) filters** can be unstable if $|a| \geq 1$, but they can achieve sharper frequency selectivity with fewer parameters.

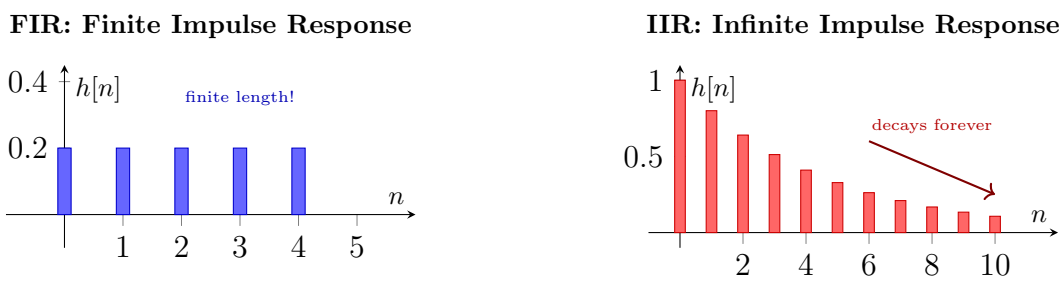

Figure 13: FIR filters have impulse responses that end after a finite number of samples (always stable). IIR filters have impulse responses that decay but never reach exactly zero (can be unstable if designed poorly).

11.9 Summary and Key Formulas

Concept	**Formula**
CT frequency response	$H(j\omega) = \int_{-\infty}^{+\infty} h(\tau)\, e^{-j\omega\tau}\, d\tau$
DT frequency response	$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n]\, e^{-j\omega n}$
CT output coefficients	$b_k = a_k \cdot H(jk\omega_0)$
DT output coefficients	$b_k = a_k \cdot H(e^{j2\pi k/N})$
RC lowpass filter	$H(j\omega) = \dfrac{1}{1 + j\omega RC}$
RC highpass filter	$G(j\omega) = \dfrac{j\omega RC}{1 + j\omega RC}$
DT first-order recursive	$H(e^{j\omega}) = \dfrac{1}{1 - a\, e^{-j\omega}}$
Ideal lowpass	$H(j\omega) = \begin{cases} 1, & \lvert\omega\rvert \le \omega_c \\ 0, & \lvert\omega\rvert > \omega_c \end{cases}$

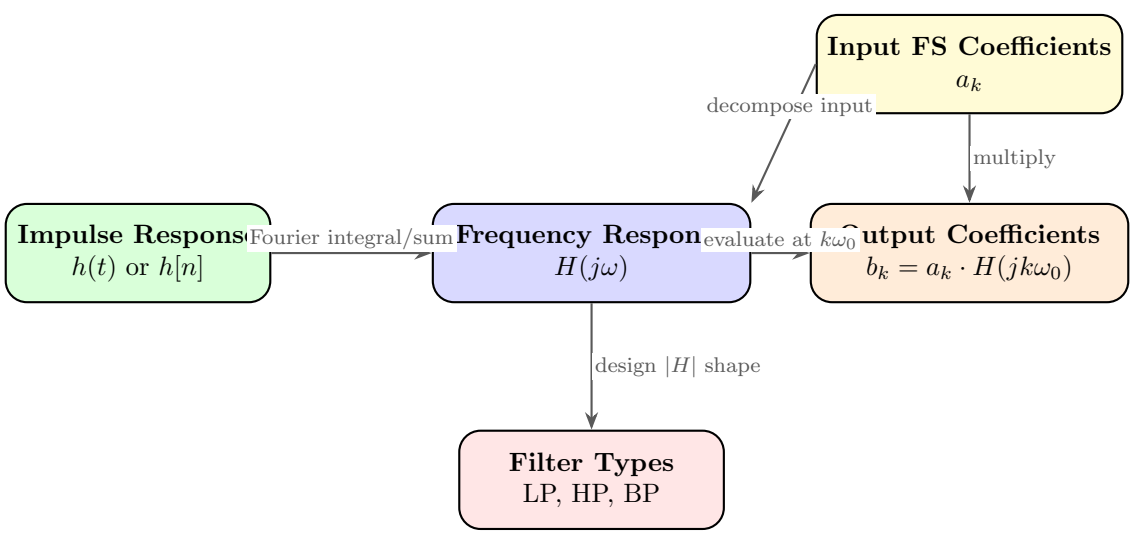

Figure 14: Concept map for Lecture 11. Everything connects through the frequency response $H(j\omega)$. The input coefficients a_k are multiplied by the frequency response values to produce the output coefficients b_k.

11.10 Common Mistakes to Avoid

1. **Evaluating $H(j\omega)$ at the wrong frequencies:** For a periodic input with fundamental frequency ω_0, you evaluate H at the **harmonic frequencies** $k\omega_0$, not at arbitrary ω. Only those specific frequencies appear in the input.

2. **Confusing the frequency response with the impulse response:** $H(j\omega)$ lives in the frequency domain; $h(t)$ lives in the time domain. They are related by an integral transform but are different functions.

3. **Forgetting DT frequency response is periodic:** $H(e^{j\omega})$ repeats every 2π. Low DT frequencies are near $\omega = 0, \pm 2\pi, \ldots$ and high DT frequencies are near $\omega = \pm\pi, \pm 3\pi, \ldots$

4. **Expecting ideal filter behavior from simple circuits:** A single RC circuit gives a gentle roll-off, not a sharp cutoff. Sharper filtering requires higher-order systems (more energy storage elements).

5. **Applying Fourier series analysis to unstable systems:** The frequency response $H(j\omega)$ is only well-defined for **stable** systems (where the impulse response is absolutely integrable). If $|a| \geq 1$ in the recursive DT filter, or if a CT system is unstable, the frequency response does not converge.

Mistake #1 Illustrated: Only evaluate H at the harmonic frequencies!

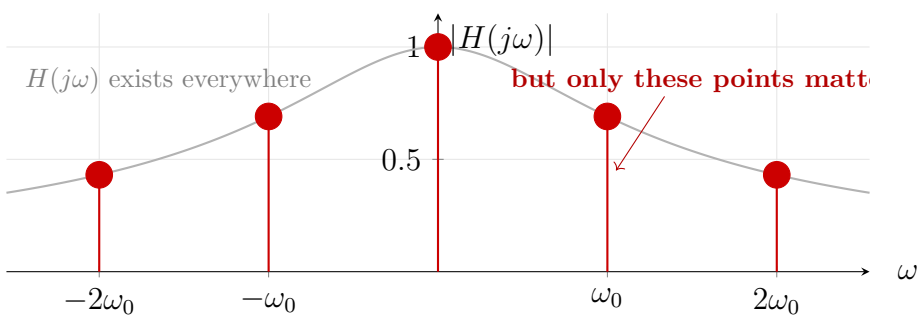

Figure 15: For a periodic input, the system only "sees" the harmonic frequencies $k\omega_0$. The red dots are the only values of $H(j\omega)$ that affect the output.

Key Insight

The central idea of this lecture: An LTI system treats a periodic input as a collection of independent frequency components and processes each one separately according to its frequency response. This decomposition into frequencies is what makes the Fourier series so powerful for understanding, designing, and analyzing LTI systems and filters.

Rogelio Gracia Otalvaro